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# COFFEQA: Counterfactual-Guided Few-Shot Distillation for LLM Question Answering

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## Abstract

Knowledge distillation aims to train a smaller student model to mimic a larger teacher model for resource-constrained environments. However, the process of distillation is typically data-intensive and computationally expensive, particularly for generative tasks. In this work, we propose *Counterfactual-guided Few-shot distillation in open-book Question Answering (COFFEQA)*, a novel strategy for distillation that aptly selects counterfactual examples for data efficiency. Counterfactuals traditionally refer to minimally perturbed instances of original data points that can change a classifier’s output. In generative tasks such as question answering (QA), given a triplet of question, context, and answer, our counterfactuals are triplets where modifying the context results in a different answer. We propose a new metric *Context Following Difficulty (CFD)* to first identify high-quality valid counterfactuals, and then leverage those counterfactual-factual pairs for distillation. We demonstrate that students distilled using COFFEQA consistently outperform those distilled only on factual data under the same number of few-shot samples. We provide theoretical guarantees on the role of counterfactuals in distillation for generative tasks by analyzing how they encourage students to be truly faithful to the teacher. Experiments across five in-domain and three out-of-domain benchmarks using two LLM families show consistent improvements over factual-only baselines, with strong generalization to both factual and counterfactual test sets. For instance, on the NaturalQuestions test set, our method improves over the baseline by 3.87 average EM points and 2.73 average F1 points under the T5 model when using 16 few-shot pairs.

## 1 Introduction

Large language models (LLMs) have achieved remarkable success across a wide range of natural language processing tasks [1–5]. However, their increasing size and computational requirements make deployment in resource-constrained environments particularly challenging [6–9]. This motivates the need for compact models with low training and inference costs that might preserve only task-relevant capabilities of larger models [10–12]. Knowledge distillation [13] seeks to address this challenge by transferring knowledge from a large teacher model to a smaller student model [3, 14, 15]. However, most distillation approaches typically rely on large-scale datasets or extensive supervision from the teacher, thus incurring prohibitive training costs [14, 16–18]. This leads to our primary question: *How can we perform data-efficient distillation in LLMs?*

Towards addressing this question, our work studies few-shot distillation for generative tasks, with a focus on open-book question answering. We propose a novel framework, *Counterfactual-guided few-shot distillation for open-book Question Answering (COFFEQA)*, a novel strategy for distillation that aptly selects counterfactual examples for data-efficient distillation in LLMs. Counterfactual-based data augmentation has shown improved performance in classification settings [19–24] by effectively learning causal dependencies and avoiding shortcut learning [25, 26]. However, their applicability

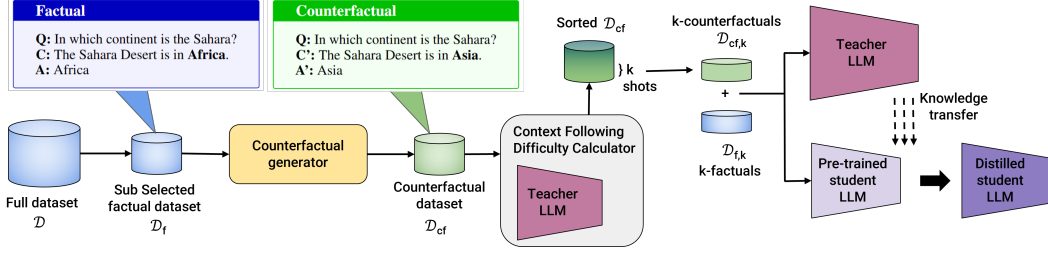


Figure 1: **Counterfactual-guided few-shot distillation framework.** We begin with the full dataset  $\mathcal{D}$  and extract samples that have entity-based answers  $\mathcal{D}_f$ . A counterfactual generator is employed to produce the counterfactual set  $\mathcal{D}_{cf}$ . Each sample is then evaluated using the teacher model via our proposed Context Following Difficulty (CFD) calculator to assess quality and alignment. Finally, the top- $k$  ranked counterfactuals are paired with their corresponding factual samples for distillation.

in few-shot distillation for generative tasks remains largely unexplored. Unlike classification tasks, generative tasks require models to produce free-form outputs, making it non-trivial to define counterfactuals. For generative tasks, a counterfactual can be viewed as a minimally modified input instruction that changes the expected output while preserving overall semantic consistency, e.g., altering a supporting fact or context in a question may lead to a different answer [25, 27–30]. We demonstrate how selecting high-quality counterfactual samples can guide knowledge transfer more effectively using extremely limited data. Our main contributions are as follows:

- **A counterfactual-guided method for few-shot distillation in generative tasks.** Our framework COFFEQA leverages counterfactual examples, constructed by minimally perturbing the input context to explicitly guide the student model toward learning causal dependencies between evidence and answers ingrained in the teacher, rather than relying solely on factual samples and the student’s pretrained knowledge. This reduces dependence on large factual training corpora and improves training efficiency under limited supervision (see Figure 1 for an overview).
- **Theoretical guarantees for leveraging counterfactuals for knowledge transfer in open-book QA.** We first introduce a notion of decision boundary in open-book QA for both the teacher and student in context-embedding space, motivated by the fact that causal variations in the input context can lead to different answers. Our formulation helps analyze distillation in open-book QA from a geometric perspective. Specifically, we show that factual-counterfactual pairs can together act as probes for the student to approximate the local geometry of the teacher’s decision boundary and become truly faithful to the teacher. To quantify the deviation between teacher and student models, we use the *Hausdorff distance* [31], which measures the worst-case deviation between their decision boundaries. We show that this deviation is bounded by counterfactual perturbation and the spread of counterfactuals in Theorem 1. Our analysis provides insights into how counterfactuals improve distillation by exposing the teacher’s boundary, thereby guiding more effective alignment.
- **Introducing Context Following Difficulty (CFD) for selecting high-quality counterfactuals.** Although we generate counterfactuals by perturbing the original input context to induce a targeted alternative answer, not all generated examples are truly valid with respect to the teacher. The teacher may not produce some target answers using the modified context or may still be able to infer them without actually relying on the context. Such cases weaken the intended causal relationship that the student was intended to learn from the teacher and may encourage shortcut learning. To address this issue, we introduce *Context Following Difficulty (CFD)*, a new metric for identifying informative counterfactuals that better capture the teacher’s ingrained behavior, thereby improving the quality of supervision during distillation.
- **Empirical validation.** We conduct comprehensive experiments on five in-domain and three out-of-domain benchmarks using the T5 [4] and FLAN-T5 [2] model families. The in-domain benchmarks include NaturalQuestions, SQuAD, NewsQA, TriviaQA, and HotpotQA, while the out-of-domain benchmarks include BioASQ, DROP, and DuoRC from the MRQA shared task [32]. We compare our method against the standard KD baseline trained on  $k$  few-shot factual samples. For a fair comparison, our COFFEQA setup uses  $k/2$  high-quality counterfactual samples selected via CFD, together with  $k/2$  factual samples. Results show that COFFEQA consistently outperforms

the baseline across datasets and  $k$ -shot settings. For instance, on the NaturalQuestions test set, our method improves over the baseline by 3.87 average EM points and 2.73 average F1 points under the T5 model when using 16 few-shot pairs (i.e., when  $k = 32$ ).

## 1.1 Related Work

**Counterfactual Data Augmentation in LLMs.** Counterfactual data augmentation has emerged as an effective approach for improving model robustness and mitigating spurious correlations in machine learning models [33–41]. Particularly for LLMs, early work relied on human-in-the-loop counterfactuals [19, 22], while later methods automated the generation process [23, 24]. Notably, Polyjuice [20] and MiCE [21] leverage LLMs to generate counterfactual examples. Subsequent work used reinforcement learning [42], active learning [43], and word dictionary-based fairness-aware pipelines [44] for counterfactual generation. While these works primarily focused on classification, counterfactual reasoning has since been extended to generative tasks such as machine translation [45], dialogue [46], and summarization [47].

**Counterfactual Question Answering.** Counterfactual reasoning has recently gained traction in question answering (QA) [48–50], particularly in open-domain settings. DisentQA [51] introduces a framework that separates parametric and contextual knowledge using counterfactual supervision, improving robustness to knowledge conflicts [52, 53]. IfQA [30] provides a benchmark for QA under hypothetical scenarios that require reasoning beyond factual retrieval. In multi-hop QA [25] and CofCA [54], counterfactual inference is formulated as a causal intervention to mitigate shortcut reasoning. Retrieval-guided methods use context-answer modifications to build counterfactual QA pairs [28, 55]. While these works demonstrate the value of counterfactual reasoning for robustness and interpretability, they primarily focus on evaluation and reasoning rather than knowledge transfer.

**Knowledge Distillation in LLMs.** Knowledge distillation [13] is a standard framework for transferring knowledge from large teacher models to smaller student models, with extensions to sequence-level distillation [15, 56–58] and task-specific distillation [14, 59, 60]. While generative and instruction-level distillation often transfer reasoning capabilities of the teacher via direct supervision or synthetic data [61, 62, 57, 17], only a few recent works [63, 64] have explored incorporating counterfactual interventions into the distillation process. However, these approaches have been primarily studied in classification settings and do not address generative tasks, where outputs are open-ended and conditioned on external knowledge.

**Our Positioning.** We extend counterfactual-based learning to **few-shot knowledge distillation in generative open-domain QA**. Unlike prior work that uses counterfactuals for robustness, reasoning, or evaluation, we leverage counterfactual QA pairs as a **structured supervision signal for knowledge transfer**, explicitly modeling conflicting knowledge during distillation.

## 2 Preliminaries

**Knowledge Distillation (KD) for Generative Tasks.** In the generative KD setting, the student learns to approximate the teacher’s conditional token distribution at each decoding step [56]. Let  $x$  denote the input and  $y = (y_1, \dots, y_T)$  denote the output sequence,  $p_s, p_t$  denote the student and teacher model distributions. The student model is trained using standard maximum likelihood estimation (MLE), given by  $\mathcal{L}_{\text{CE}} = -\sum_{t=1}^T \log p_s(y_t \mid x, y_{<t})$ . In addition to the MLE objective, KD introduces a *distillation loss* that aligns the student’s token-level predictions with those of the teacher. Specifically, at each decoding step, the Kullback-Leibler (KL) divergence between the teacher and student distributions is calculated as  $\mathcal{L}_{\text{KD}} = \sum_{t=1}^T \text{KL}(p_t(\cdot \mid x, y_{<t}) \parallel p_s(\cdot \mid x, y_{<t}))$ . The final training objective is a weighted combination of the MLE loss and the distillation loss  $\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) \mathcal{L}_{\text{KD}}$ , where  $\alpha \in [0, 1]$  controls the trade-off between fitting the ground-truth sequence and matching the teacher’s distribution.

**Instruction Following Difficulty.** Instruction Following Difficulty (IFD) measures how much useful information an instruction provides for generating the corresponding response [65]. Intuitively, a lower IFD score indicates that the instruction effectively guides the model in generating the answer, whereas a higher IFD score indicates less instructional help or greater difficulty with instructions. For open-domain question answering, we consider triplets  $(Q, C, A)$ , where  $Q$  is the question,  $C$  is the context, and  $A = (a_1, \dots, a_N)$  is the answer sequence. Then, the Instruction Following Difficulty

| Factual Sample                                                                                                                                                                                                                                                                                                       | Counterfactual Sample                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Q:</b> In which European country is the site of the Battle of Marengo?</p> <p><b>C:</b> [DOC] [TLE] ... Battle of Marengo, June 14, 1800 ... A map of Marengo in northern <b>Italy</b>, site of the important battle ... It secured victory for Napoleon's <b>Italy</b> campaign...</p> <p><b>A:</b> Italy</p> | <p><b>Q:</b> In which European country is the site of the Battle of Marengo?</p> <p><b>C':</b> [DOC] [TLE] ... Battle of Marengo, June 14, 1800 ... A map of Marengo in northern <b>Turkey</b>, site of the important battle ... It secured victory for Napoleon's <b>Turkey</b> campaign...</p> <p><b>A':</b> Turkey</p> |

Figure 2: Example factual and counterfactual samples used in COFFEQA. The counterfactual sample is generated by minimally perturbing entity-related context terms while changing the target answer.

(IFD) is defined below, where  $L_\theta$  denotes the cross-entropy loss given the model parameterized by  $\theta$ .

$$\text{IFD}_\theta(Q, C, A) = \frac{L_\theta(A | Q, C)}{L_\theta(A)} = \frac{-\frac{1}{N} \sum_{i=1}^N \log p_\theta(a_i | Q, C, a_{<i})}{-\frac{1}{N} \sum_{i=1}^N \log p_\theta(a_i | a_{<i})} \quad (1)$$

**Counterfactual Definition and Generation.** We define counterfactual generation for open-book QA as constructing minimally modified training examples that preserve the original question while changing the evidence to support a different answer. Given a triplet  $(Q, C, A)$ , where  $Q$  is the question,  $C$  is the context, and  $A$  is the answer, the counterfactual generation framework constructs a counterfactual example by modifying the context using an alternative answer. The resulting counterfactual triplet is denoted as  $(Q, C', A')$ . First, the samples with named entity answers are filtered out by considering five named entity categories: *person*, *date*, *numeric*, *organization*, and *location*. We assume that the training set contains a sufficient number of samples with named entity answers to support this filtering. Next, an alternative answer  $A'$  is randomly sampled from the answer distribution of the dataset which has the same entity type as  $A$ . The original answer  $A$  is then replaced with  $A'$  in the context  $C$  to construct  $C'$ . This process results in a modified example where the context is altered while preserving the structure of the original question. An example of a factual and its corresponding counterfactual sample is in Figure 2. We use the counterfactual generation framework from [27] which mitigates knowledge conflicts in open-book QA models.

### 3 Main Contributions

#### 3.1 Theoretical guarantees of leveraging counterfactuals for distillation in open-book QA

To demonstrate the role of counterfactuals in distillation, we first introduce a notion of decision boundary for QA tasks. While classification tasks define a “*label-flipping boundary*”, for open-book QA we define a “*context-response boundary*” in the context embedding space. In classification, the boundary corresponds to the regions where two classes have equal probabilities. In contrast, in open-book QA, it represents the transition region in context embedding space where the model shifts from generating answer  $A$  to its counterfactual answer  $A'$ . We interpret these regions as two subspaces within the context embedding space, denoted by  $\mathcal{A}$  and  $\mathcal{A}'$ , respectively. However, since generative tasks are inherently open-ended, there may also exist a third region where the model produces an answer  $A''$ , different from both  $A$  and  $A'$  corresponding to out-of-context or unrelated outputs. In this section, we assume that the model perfectly generates either  $A$  or  $A'$  depending on the given context. After defining decision boundaries, our goal is to show that using factual-counterfactual pairs during distillation brings the student boundary much closer to the teacher’s boundary. We use the Hausdorff distance [31] to measure the worst-case deviation between the teacher and student boundaries, and derive an upper bound for this deviation.

**Definition of Hausdorff Distance [66].** Let  $(\mathcal{M}, d)$  be a metric space. For each pair of non-empty subsets  $\mathcal{M}_t \subset \mathcal{M}$  and  $\mathcal{M}_s \subset \mathcal{M}$ , the Hausdorff distance between  $\mathcal{M}_t$  and  $\mathcal{M}_s$  is defined as:

$$d_H(\mathcal{M}_t, \mathcal{M}_s) := \max \left\{ \sup_{c_t \in \mathcal{M}_t} \inf_{c_s \in \mathcal{M}_s} d(c_t, c_s), \sup_{c_s \in \mathcal{M}_s} \inf_{c_t \in \mathcal{M}_t} d(c_t, c_s) \right\} \quad (2)$$

This measures the maximum of the minimum distances between two sets, capturing the worst-case geometric deviation. (see Figure 3).

**Theorem 1** Let  $\mathcal{C}$  be the space of context embeddings,  $c \in \mathcal{C}$  arbitrary input context,  $Q \in \mathcal{Q}$  input question,  $A \in \mathcal{A}$  factual answer, and  $A' \in \mathcal{A}'$  counterfactual answer. Consider teacher and student models with response functions  $G_t(Q, c) = \arg \max_A P_t(A|Q, c)$  and  $G_s(Q, c) = \arg \max_{A'} P_s(A'|Q, c)$ . The respective context-switching boundaries can be defined as the sets:

$$\mathcal{M}_t = \{c \in \mathcal{C} \mid P_t(A|Q, c) = P_t(A'|Q, c)\}, \quad \mathcal{M}_s = \{c \in \mathcal{C} \mid P_s(A|Q, c) = P_s(A'|Q, c)\}.$$

Suppose there exists a factual-counterfactual paired dataset  $\mathcal{D}_{c+cf} = \{(c_i, c'_i)\}_{i=1}^k$  for the above  $\mathcal{M}_t$  that satisfies three conditions: (1) Minimal context perturbation in context embedding space is bounded by  $\alpha$  (i.e.,  $\|c_i - c'_i\| \leq \alpha$ ), (2) Exact answer generation:  $G_t(Q, c) = G_s(Q, c)$  where  $c = \{c_i, c'_i\}$ , (3)  $\epsilon$ -spread context-sensitivity space: counterfactuals are well-distributed across the context-sensitivity space with spread  $\epsilon$ . i.e., if  $t \in \mathcal{M}_t$ , then  $\exists t_i^*$  such that  $\|t - t_i^*\| \leq \epsilon$ . This applies for  $\mathcal{M}_s$  as well. i.e., if  $s \in \mathcal{M}_s$ , then  $\exists s_i^*$  such that  $\|s - s_i^*\| \leq \epsilon$ .

Then, the Hausdorff distance between the teacher’s and student’s context-response boundaries satisfies:

$$d_H(\mathcal{M}_t, \mathcal{M}_s) \leq \alpha + \epsilon.$$

The proof can be found in Appendix C.

**Intuition behind theorem and assumptions.** The core idea of the theorem is that the student model will closely mirror the teacher model’s behavior if three conditions are satisfied. The first condition enforces that the maximum perturbation of any sample is bounded by  $\alpha$ . In the open-book QA setting, counterfactuals are minimally perturbed versions of the input context (e.g., changing “The CEO is John” to “The CEO is Mark”) while preserving the remaining structural semantics. Therefore, perturbations are inherently small by construction, making  $\alpha$  small. In the second condition, we assume the student is a focused learner. If the teacher says the answer to the original question is “John” and the answer to the counterfactual question is “Mark,” the student’s training loss (the KL loss) forces it to match those specific predictions exactly. The third condition assumes that the counterfactual pairs are well distributed.

Intuitively, this ensures that no region of the teacher or student space is left unexplored so that every point on the decision boundary is within a distance of at most  $\epsilon$  from some sampled counterfactual pair cut point in the boundary. If we have, well-spread counterfactual pairs  $\epsilon$  become small. Together, small  $\alpha$  and small  $\epsilon$  guarantee that the student boundary remains inside an small  $(\alpha + \epsilon)$ -tube around the teacher boundary showing the advantage of using factual-counterfactual pairs for distillation.

However, for this theorem to hold, the teacher must exactly generate the counterfactual answer  $A'$  given the counterfactual context  $C'$ . This is not guaranteed due to inherently stochastic and unpredictable generative behavior of LLMs [67]. To address this limitation and identify reliable counterfactuals, we introduce a new selection criterion in the following section.

### 3.2 Introducing Context Following Difficulty (CFD) for selecting counterfactuals

Ideally, generated counterfactuals would be evaluated using the teacher model to verify whether they exactly produce the target response  $A'$ . However, in generative tasks, exact matching is not guaranteed due to the stochastic and uncontrolled nature of generative LLMs [67]. In particular, the teacher may produce additional tokens, paraphrases, or aliases of the expected answer, making strict equivalence checking unreliable. Therefore, we draw inspiration from the Instruction Following Difficulty (IFD) metric [65] to check the relevance of the input context towards generating the expected answer. Our proposed Context Following Difficulty (CFD) is defined as:

$$CFD = \frac{L_\theta(A'|Q, C')}{L_\theta(A'|Q)} \quad (3)$$

We contend that counterfactuals with the lowest CFD score are the most informative and valid samples for distillation that help mimic the teacher’s boundary. If the teacher model is given  $Q$  with

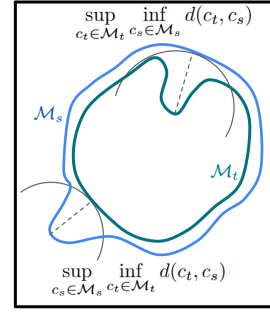


Figure 3: Visualizing the Hausdorff distance between  $\mathcal{M}_t$  and  $\mathcal{M}_s$ .

context  $C'$ , the cross-entropy loss  $L_\theta(A' | Q, C')$  is low if the teacher correctly generates or closely approximates  $A'$ . However, this measurement alone is not sufficient, since some  $A'$  instances may be easily predictable from  $Q$  without requiring the counterfactual context. Therefore, we aim to retain samples for which  $A'$  is generated only when  $C'$  is provided along with  $Q$ . In other words, we prefer samples where the teacher does not generate  $A'$  from  $Q$  alone, i.e., where  $L_\theta(A' | Q)$  is high. These two requirements are precisely captured by the CFD metric when we select the samples with the lowest CFD values.

Interestingly, CFD also helps filter out “wrong” or “invalid” counterfactuals by capturing out-of-context responses that fall into the region  $A''$ . Simply targeting an alternative answer  $A'$  does not guarantee a successful transition within the teacher’s manifold. In our counterfactual generation process, we target a specific counterfactual answer  $A'$  to produce a perturbed context  $C'$ . However, we cannot guarantee that feeding  $C'$  into the teacher model will exactly produce  $A'$ . As illustrated in Figure 4, in some cases, the counterfactual generation framework may generate a counterfactual sample  $(Q_a, C'_a)$  from  $(Q_a, C_a)$  that leads to a garbage answer  $A''_a$  entirely different from the targeted answer  $A'_a$ . In such cases, the loss  $L_\theta(A'_a | Q_a, C'_a)$  remains high, resulting in a high CFD value. Similarly, for a sample  $(Q_b, C_b)$ , the generated counterfactual  $(Q_b, C'_b)$  may still yield the original answer  $A_b$ , indicating that the intervention has failed to induce the desired transition. This also leads to a high CFD value, since  $L_\theta(A'_b | Q_b, C'_b)$  remains high. In contrast, we aim to select samples such as  $(Q_c, C'_c)$ , where the generated counterfactual exhibits a successful and meaningful transition aligned with the target behavior, resulting in low CFD values.

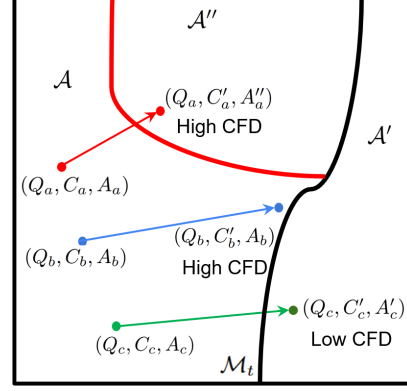


Figure 4: CFD-based counterfactual evaluation. Generated counterfactuals  $(Q_a, C'_a)$  and  $(Q_b, C'_b)$  fail to cross the teacher decision boundary  $M_t$ , resulting in a high CFD value. In contrast, sample  $(Q_c, C'_c)$  successfully crosses the boundary to induce the target answer  $A'$ , yielding a low CFD.

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#### Algorithm 1 COFFEQA

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**Input:**  $\mathcal{D}_f$ , few-shot size  $k$ , teacher  $G_t$ , student  $G_s$ , epochs  $E$ , learning rate  $\eta$ , weight  $\alpha$

- 1:  $\mathcal{D}_{cf} \leftarrow \emptyset$
- 2: **for all**  $(Q, C, A) \in \mathcal{D}_f$  **do**
- 3:    $(Q, C', A') \leftarrow \text{GenCF}(Q, C, A)$
- 4:    $\mathcal{D}_{cf} \leftarrow \mathcal{D}_{cf} \cup \{(Q, C', A')\}$
- 5: **end for**
- 6: **for all**  $(Q, C', A') \in \mathcal{D}_{cf}$  **do**
- 7:   Compute  $\text{CFD}((Q, C', A'); G_t)$
- 8: **end for**
- 9: Select lowest CFD  $k$  counterfactual samples  $\mathcal{D}_{k,cf} \subset \mathcal{D}_{cf}$
- 10: Find corresponding factials  $\mathcal{D}_{k,f} \subset \mathcal{D}_f$
- 11:  $\mathcal{D}_{\text{KD}} \leftarrow \mathcal{D}_{k,f}, \mathcal{D}_{\text{COFFEQA}} \leftarrow \mathcal{D}_{k/2,f} \cup \mathcal{D}_{k/2,cf}$
- 12: **for all**  $\mathcal{D}_{\text{train}} \in \{\mathcal{D}_{\text{KD}}, \mathcal{D}_{\text{COFFEQA}}\}$  **do**
- 13:   **for**  $e = 1$  **to**  $E$  **do**
- 14:     **for all**  $(Q, C, A) \in \mathcal{D}_{\text{train}}$  **do**
- 15:        $\mathcal{L}_{\text{CE}} \leftarrow \text{CE}(G_s(Q, C), A)$
- 16:        $\mathcal{L}_{\text{KD}} \leftarrow \text{KL}(G_t(Q, C) \parallel G_s(Q, C))$
- 17:        $\mathcal{L} \leftarrow \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) \mathcal{L}_{\text{KD}}$
- 18:       Update  $\theta_s \leftarrow \theta_s - \eta \nabla_{\theta_s} \mathcal{L}$
- 19:     **end for**
- 20:   **end for**
- 21: **return** distilled student  $G_s$
- 22: **end for**

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Therefore, after counterfactual generation, we select the  $k$  counterfactual samples with the lowest CFD values, along with their corresponding factual pairs, for our experiments. This is done under the assumption that the full dataset is available during distillation, with the goal of improving both data efficiency and computational efficiency.

**COFFEQA Algorithm.** Refer to Algorithm 1 and Figure 1. Consider, we have the fine-tuned teacher  $G_t$  trained on the whole dataset  $\mathcal{D}$ . From that, we sample subset  $\mathcal{D}_f$  consisting of samples with entity-based answers such as *person*, *date*, *numeric*, *organization* and *location* entities. Next, we generate  $\mathcal{D}_{cf}$  using the counterfactual generation framework. These counterfactuals are then evaluated using the CFD metric, and the  $k$  counterfactuals with the lowest CFD are selected. For distillation, we define two training sets. The baseline training set  $\mathcal{D}_{\text{KD}}$  consists of corresponding factual counterparts corresponding to the selected counterfactual samples in  $\mathcal{D}_{cf}$ . In contrast, our proposed COFFEQA training set  $\mathcal{D}_{\text{COFFEQA}}$  consists of the lowest CFD  $k/2$  counterfactuals paired with their corresponding factials. Finally, the distilled student  $G_s$  is evaluated on the test dataset.

Table 1: Performance comparison on in-domain benchmarks using the T5 model. Bold values indicate the better performing method (KD or COFFEQA). Blue shading highlights aggregate results.

| Dataset  | Score | Method  | Total Samples ( $k$ )    |                          |                          |                          |                          | Avg $\pm$ std           |
|----------|-------|---------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------|
|          |       |         | 8                        | 16                       | 32                       | 64                       | 128                      |                         |
| NQ       | EM    | KD      | <b>56.12</b> $\pm$ 1.81  | 55.30 $\pm$ 1.62         | 52.89 $\pm$ 4.29         | <b>55.36</b> $\pm$ 1.23  | 54.07 $\pm$ 4.60         | 54.75 $\pm$ 1.27        |
|          |       | COFFEQA | 54.09 $\pm$ 2.84         | <b>57.29</b> $\pm$ 1.24  | <b>56.76</b> $\pm$ 0.67  | 55.23 $\pm$ 2.56         | <b>54.18</b> $\pm$ 4.20  | <b>55.51</b> $\pm$ 1.47 |
|          | F1    | KD      | <b>61.09</b> $\pm$ 0.93  | 60.14 $\pm$ 0.95         | 58.39 $\pm$ 3.47         | 60.08 $\pm$ 0.87         | 58.69 $\pm$ 3.97         | 59.68 $\pm$ 1.12        |
|          |       | COFFEQA | 59.37 $\pm$ 2.04         | <b>61.82</b> $\pm$ 0.99  | <b>61.12</b> $\pm$ 0.78  | <b>60.17</b> $\pm$ 2.34  | <b>58.93</b> $\pm$ 3.18  | <b>60.28</b> $\pm$ 1.20 |
| SQuAD    | EM    | KD      | 78.82 $\pm$ 0.24         | 78.70 $\pm$ 0.30         | 78.48 $\pm$ 0.24         | 77.28 $\pm$ 2.02         | 77.50 $\pm$ 0.51         | 78.16 $\pm$ 0.71        |
|          |       | COFFEQA | <b>79.06</b> $\pm$ 0.15  | <b>78.88</b> $\pm$ 0.44  | <b>78.89</b> $\pm$ 0.37  | <b>77.99</b> $\pm$ 0.65  | <b>77.58</b> $\pm$ 0.97  | <b>78.48</b> $\pm$ 0.65 |
|          | F1    | KD      | 87.04 $\pm$ 0.20         | 86.95 $\pm$ 0.18         | 86.91 $\pm$ 0.13         | 85.46 $\pm$ 1.98         | 85.81 $\pm$ 0.57         | 86.43 $\pm$ 0.74        |
|          |       | COFFEQA | <b>87.16</b> $\pm$ 0.15  | <b>87.00</b> $\pm$ 0.39  | <b>86.95</b> $\pm$ 0.24  | <b>86.22</b> $\pm$ 0.54  | <b>85.84</b> $\pm$ 0.45  | <b>86.63</b> $\pm$ 0.57 |
| NewsQA   | EM    | KD      | 62.81 $\pm$ 2.79         | 64.99 $\pm$ 2.21         | <b>68.81</b> $\pm$ 0.85  | <b>71.19</b> $\pm$ 1.28  | <b>72.41</b> $\pm$ 1.49  | <b>68.04</b> $\pm$ 4.07 |
|          |       | COFFEQA | <b>66.06</b> $\pm$ 0.85  | <b>67.94</b> $\pm$ 2.12  | 67.04 $\pm$ 1.40         | 68.70 $\pm$ 1.60         | 70.20 $\pm$ 1.02         | 67.99 $\pm$ 1.58        |
|          | F1    | KD      | 70.73 $\pm$ 2.94         | 73.03 $\pm$ 1.79         | <b>75.57</b> $\pm$ 1.15  | <b>77.05</b> $\pm$ 0.87  | <b>78.27</b> $\pm$ 1.25  | <b>74.93</b> $\pm$ 3.05 |
|          |       | COFFEQA | <b>74.41</b> $\pm$ 0.73  | <b>74.86</b> $\pm$ 1.39  | 74.01 $\pm$ 1.05         | 75.20 $\pm$ 1.52         | 76.08 $\pm$ 0.77         | 74.91 $\pm$ 0.79        |
| TriviaQA | EM    | KD      | <b>57.43</b> $\pm$ 0.52  | 56.54 $\pm$ 0.89         | 56.51 $\pm$ 1.03         | 58.02 $\pm$ 2.90         | 57.33 $\pm$ 1.44         | 57.17 $\pm$ 0.64        |
|          |       | COFFEQA | 56.31 $\pm$ 0.64         | <b>57.20</b> $\pm$ 0.68  | <b>58.75</b> $\pm$ 2.08  | <b>59.90</b> $\pm$ 0.85  | <b>59.44</b> $\pm$ 0.83  | <b>58.32</b> $\pm$ 1.52 |
|          | F1    | KD      | <b>63.57</b> $\pm$ 0.51  | 62.69 $\pm$ 0.55         | 62.62 $\pm$ 0.42         | 62.20 $\pm$ 2.83         | 60.51 $\pm$ 1.43         | 62.32 $\pm$ 1.13        |
|          |       | COFFEQA | 62.65 $\pm$ 0.34         | <b>63.07</b> $\pm$ 0.58  | <b>63.41</b> $\pm$ 2.01  | <b>64.06</b> $\pm$ 0.52  | <b>63.07</b> $\pm$ 0.77  | <b>63.25</b> $\pm$ 0.53 |
| HotpotQA | EM    | KD      | 62.54 $\pm$ 0.93         | 63.68 $\pm$ 1.98         | 62.25 $\pm$ 1.12         | 63.28 $\pm$ 1.60         | <b>64.70</b> $\pm$ 1.73  | 63.29 $\pm$ 0.97        |
|          |       | COFFEQA | <b>64.53</b> $\pm$ 0.79  | <b>64.48</b> $\pm$ 1.22  | <b>63.59</b> $\pm$ 1.54  | <b>63.68</b> $\pm$ 1.41  | 62.49 $\pm$ 1.93         | <b>63.75</b> $\pm$ 0.83 |
|          | F1    | KD      | 72.26 $\pm$ 0.69         | 73.44 $\pm$ 1.63         | <b>73.15</b> $\pm$ 1.76  | <b>73.14</b> $\pm$ 2.40  | <b>74.45</b> $\pm$ 1.20  | <b>73.29</b> $\pm$ 0.79 |
|          |       | COFFEQA | <b>73.85</b> $\pm$ 0.63  | <b>73.81</b> $\pm$ 0.92  | 72.24 $\pm$ 0.81         | 73.08 $\pm$ 1.35         | 71.32 $\pm$ 2.05         | 72.86 $\pm$ 1.08        |
| AVG      | EM    | KD      | 63.54 $\pm$ 9.05         | 63.84 $\pm$ 9.33         | 63.79 $\pm$ 10.19        | 65.03 $\pm$ 9.13         | <b>65.20</b> $\pm$ 9.87  | 64.28 $\pm$ 0.77        |
|          |       | COFFEQA | <b>64.01</b> $\pm$ 9.86  | <b>65.16</b> $\pm$ 8.97  | <b>65.01</b> $\pm$ 8.75  | <b>65.10</b> $\pm$ 8.74  | 64.78 $\pm$ 9.21         | <b>64.81</b> $\pm$ 0.47 |
|          | F1    | KD      | 70.94 $\pm$ 10.15        | 71.25 $\pm$ 10.62        | 71.33 $\pm$ 11.26        | 71.59 $\pm$ 10.55        | <b>71.55</b> $\pm$ 11.66 | 71.33 $\pm$ 0.26        |
|          |       | COFFEQA | <b>71.49</b> $\pm$ 11.01 | <b>72.11</b> $\pm$ 10.24 | <b>71.55</b> $\pm$ 10.23 | <b>71.75</b> $\pm$ 10.20 | 71.05 $\pm$ 10.66        | <b>71.59</b> $\pm$ 0.39 |

## 4 Experiments

**Datasets.** We evaluate COFFEQA on eight open-domain question answering benchmarks from the MRQA 2019 shared task [32], comprising five in-domain and three out-of-domain datasets. The in-domain benchmarks include Natural Questions (NQ), SQuAD, NewsQA, TriviaQA, and HotpotQA. NQ consists of real user queries collected from Google search logs [68]. SQuAD (Stanford Question Answering Dataset) contains questions authored by crowdworkers based on Wikipedia articles [69]. NewsQA is constructed from CNN news articles with questions generated crowdworkers [70]. TriviaQA consists QA pairs sourced from trivia and quiz-league websites [71]. HotpotQA focuses on multi-hop reasoning, requiring models to combine information from multiple Wikipedia articles [72]. The out-of-domain benchmarks include BioASQ, DROP, and DuoRC. BioASQ is a large-scale biomedical QA dataset created by domain experts [73]. DROP contains quantitative QA pairs which has numerical and text answers [74]. DuoRC contains QA pairs derived from movie plots, constructed by crowdworkers using IMDb and Wikipedia [75]. (see Appendix B.1 for additional details.)

**Models.** We evaluate our method using two model families: T5 [4] and FLAN-T5 [2]. In the T5 setting, T5-Large (770M parameters) serves as the teacher model and T5-Base (220M) as the student model. In the FLAN-T5 setting, FLAN-T5-XL (3B parameters) serves as the teacher and FLAN-T5-Base (250M) is used as the student. Full training details are provided in Appendix B.2.

**Setup:** By following the prior work [63], we first fine-tune the teacher model on the full training dataset. We then generate counterfactuals for each factual sample and evaluate their quality using the CFD metric. Based on the CFD scores, counterfactuals are ranked, and we select the top- $k \in \{8, 16, 32, 64, 128\}$  lowest CFD counterfactuals along with their corresponding factual pairs. For the baseline, we perform distillation using only those selected  $k$  factual samples. For our COFFEQA setting, we select the top  $k/2$  lowest CFD counterfactuals along with corresponding  $k/2$  factual samples. We evaluate performance using Exact Match (EM) and F1 score on the test sets, reporting the mean and standard deviation over five random seeds. Results for the T5 model family are presented in Table 1. Results for the FLAN-T5 model family are shown in Table 3. Results are averaged across different few-shot settings ( $k$ -shots) for each dataset, as well as across datasets for a fixed  $k$  (reported in the last rows and final columns) to identify overall performance.

Table 2: Performance comparison on out-of-domain benchmarks using the T5 model. Bold values indicate the better performing method (KD or COFFEQA). Blue shading highlights aggregate results.

| Dataset | Score | Method  | Total Samples ( $k$ )   |                         |                         |                         |                         | Avg $\pm$ std           |
|---------|-------|---------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|         |       |         | 8                       | 16                      | 32                      | 64                      | 128                     |                         |
| BioASQ  | EM    | KD      | <b>26.14</b> $\pm$ 2.27 | <b>26.67</b> $\pm$ 2.37 | 25.79 $\pm$ 1.00        | <b>27.19</b> $\pm$ 3.51 | 26.49 $\pm$ 2.27        | <b>26.46</b> $\pm$ 0.53 |
|         |       | COFFEQA | 23.16 $\pm$ 3.54        | 26.14 $\pm$ 2.51        | <b>26.84</b> $\pm$ 2.02 | 26.49 $\pm$ 2.35        | <b>28.42</b> $\pm$ 2.29 | 26.21 $\pm$ 1.91        |
|         | F1    | KD      | <b>38.52</b> $\pm$ 1.04 | 38.06 $\pm$ 1.81        | 37.90 $\pm$ 1.20        | <b>38.93</b> $\pm$ 3.37 | 41.28 $\pm$ 3.63        | 38.94 $\pm$ 1.37        |
|         |       | COFFEQA | 36.53 $\pm$ 2.28        | <b>39.18</b> $\pm$ 2.13 | <b>39.00</b> $\pm$ 1.92 | 38.58 $\pm$ 3.01        | <b>41.61</b> $\pm$ 2.93 | <b>38.98</b> $\pm$ 1.81 |
| DROP    | EM    | KD      | 33.82 $\pm$ 1.25        | 32.28 $\pm$ 0.69        | 31.31 $\pm$ 1.74        | <b>32.57</b> $\pm$ 0.83 | 30.91 $\pm$ 1.76        | 32.18 $\pm$ 1.14        |
|         |       | COFFEQA | <b>33.98</b> $\pm$ 0.91 | <b>33.45</b> $\pm$ 0.85 | <b>33.41</b> $\pm$ 0.47 | 30.75 $\pm$ 1.48        | <b>31.96</b> $\pm$ 1.95 | <b>32.71</b> $\pm$ 1.33 |
|         | F1    | KD      | 38.12 $\pm$ 1.31        | 36.40 $\pm$ 0.73        | <b>37.77</b> $\pm$ 0.49 | 35.02 $\pm$ 1.29        | 35.68 $\pm$ 1.75        | 36.60 $\pm$ 1.33        |
|         |       | COFFEQA | <b>38.36</b> $\pm$ 0.68 | <b>37.83</b> $\pm$ 0.84 | 35.20 $\pm$ 1.99        | <b>36.58</b> $\pm$ 0.78 | <b>36.76</b> $\pm$ 2.34 | <b>36.95</b> $\pm$ 1.22 |
| DuoRC   | EM    | KD      | 41.87 $\pm$ 1.43        | 39.38 $\pm$ 1.18        | <b>39.87</b> $\pm$ 2.05 | 37.68 $\pm$ 2.18        | 35.55 $\pm$ 2.45        | 38.87 $\pm$ 2.38        |
|         |       | COFFEQA | <b>42.18</b> $\pm$ 0.43 | <b>39.82</b> $\pm$ 1.72 | 36.93 $\pm$ 2.72        | <b>40.58</b> $\pm$ 1.90 | <b>37.37</b> $\pm$ 1.68 | <b>39.38</b> $\pm$ 2.21 |
|         | F1    | KD      | 49.11 $\pm$ 1.04        | 47.29 $\pm$ 1.19        | 45.23 $\pm$ 2.89        | <b>48.08</b> $\pm$ 1.81 | 44.97 $\pm$ 1.69        | 46.94 $\pm$ 1.80        |
|         |       | COFFEQA | <b>49.48</b> $\pm$ 0.46 | <b>48.20</b> $\pm$ 1.29 | <b>47.82</b> $\pm$ 1.55 | 46.34 $\pm$ 2.15        | <b>45.86</b> $\pm$ 1.21 | <b>47.54</b> $\pm$ 1.46 |
| AVG     | EM    | KD      | <b>33.94</b> $\pm$ 7.87 | 32.78 $\pm$ 6.37        | 32.32 $\pm$ 7.09        | 32.48 $\pm$ 5.25        | 30.98 $\pm$ 4.53        | 32.50 $\pm$ 1.06        |
|         |       | COFFEQA | 33.11 $\pm$ 9.54        | <b>33.14</b> $\pm$ 6.85 | <b>32.39</b> $\pm$ 5.12 | <b>32.61</b> $\pm$ 7.23 | <b>32.58</b> $\pm$ 4.51 | <b>32.77</b> $\pm$ 0.34 |
|         | F1    | KD      | <b>41.92</b> $\pm$ 6.23 | 40.58 $\pm$ 5.87        | 40.30 $\pm$ 4.27        | <b>40.68</b> $\pm$ 6.70 | 40.64 $\pm$ 4.68        | 40.82 $\pm$ 0.63        |
|         |       | COFFEQA | 41.46 $\pm$ 7.01        | <b>41.74</b> $\pm$ 5.64 | <b>40.67</b> $\pm$ 6.47 | 40.50 $\pm$ 5.16        | <b>41.41</b> $\pm$ 4.55 | <b>41.16</b> $\pm$ 0.54 |

#### 4.1 Main Experimental Results

**COFFEQA outperforms the baseline.** As demonstrated in Table 1, the proposed method consistently outperforms the standard KD baseline in the few-shot setting. Specifically, COFFEQA achieves an overall average EM score of 64.81, compared to 64.28 for the baseline, across all benchmarks and  $k$ -shot settings. It also attains a higher average F1 score of 71.59, compared to the baseline 71.33. Furthermore, for the T5 model family, our method outperforms the baseline on four out of five benchmarks and achieves comparable performance on the remaining NewsQA benchmark. We also observed that COFFEQA exhibits more stable performance across different  $k$ -shot settings within each benchmark, as reflected by its lower standard deviation in most cases (see the last column of Table 1). A similar trend is observed for the FLAN-T5 model family. As shown in Table 3, COFFEQA consistently outperforms the baseline across all benchmarks and  $k$ -shot settings (see the last column values of the final row).

**COFFEQA generalizes well on out-of-domain datasets.** To examine the effect of domain bias, we conduct experiments on out-of-domain datasets. Specifically, we use the Natural Questions dataset for distillation and evaluated the resulting student models on three out-of-domain test sets. As shown in Table 2, for the T5 model family COFFEQA outperforms standard KD when averaged across the datasets and  $k$ -shots. In particular, our method achieves an overall EM score of 32.77, compared 32.50 for KD. Also, it has a higher F1 score of 41.16, compared to 40.82 for KD. A similar trend is observed for the FLAN-T5 model family. Refer to Table 8 in Appendix E for more details.

**Our method performs well across different model families.** We conducted experiments on two model variants, T5 and FLAN-T5. In the T5 setting, the teacher is  $3.5\times$  larger than the student, while in FLAN-T5, the teacher is  $12\times$  larger than the student. Overall, COFFEQA outperforms the baseline in both model families. We observe that the performance gains of COFFEQA are particularly strong in low- $k$  settings, highlighting the effectiveness of the proposed approach in the few-shot regime. As  $k$  increases, the gap between KD and COFFEQA gradually decreases, and at  $k = 128$ , the KD baseline surpasses COFFEQA in both model families (see the last row of Table 1 and 3).

#### 4.2 Ablation Study

**Our method is robust to knowledge conflict.** We generate counterfactuals for each test set and evaluate model performance to analyze how the distilled students handle knowledge conflicts. In factual data, some questions can be answered using pre-trained knowledge, while others require reliance on the provided context. In contrast, counterfactual examples are substantially more challenging since the original answer, which is associated with pre-trained knowledge, has already been altered in the context. As a result, models must rely less on memorized knowledge and more

Table 3: Performance comparison on in-domain benchmarks using the FLAN-T5 model. Bold values indicate the better performing method (KD or COFFEQA). Blue shading highlights aggregate results.

| Dataset  | Score | Method  | Total Samples ( $k$ )   |                         |                         |                         |                         | Avg $\pm$ std           |
|----------|-------|---------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|          |       |         | 8                       | 16                      | 32                      | 64                      | 128                     |                         |
| NQ       | EM    | KD      | <b>63.20</b> $\pm$ 1.11 | 63.10 $\pm$ 1.12        | 63.13 $\pm$ 2.57        | 64.08 $\pm$ 1.69        | <b>64.58</b> $\pm$ 1.22 | 63.62 $\pm$ 0.67        |
|          |       | COFFEQA | 62.92 $\pm$ 1.55        | <b>63.92</b> $\pm$ 0.62 | <b>63.66</b> $\pm$ 0.73 | <b>65.26</b> $\pm$ 1.20 | 64.42 $\pm$ 1.27        | <b>64.04</b> $\pm$ 0.87 |
|          | F1    | KD      | <b>67.91</b> $\pm$ 0.89 | 68.10 $\pm$ 0.87        | 68.30 $\pm$ 2.13        | <b>70.11</b> $\pm$ 0.72 | 69.14 $\pm$ 1.23        | 68.71 $\pm$ 0.91        |
|          |       | COFFEQA | 67.82 $\pm$ 1.29        | <b>68.53</b> $\pm$ 0.55 | <b>69.02</b> $\pm$ 0.68 | 68.94 $\pm$ 1.72        | <b>69.77</b> $\pm$ 0.74 | <b>68.82</b> $\pm$ 0.71 |
| SQuAD    | EM    | KD      | 78.98 $\pm$ 0.14        | <b>79.84</b> $\pm$ 0.40 | 79.66 $\pm$ 0.48        | 80.14 $\pm$ 0.45        | <b>80.24</b> $\pm$ 0.48 | <b>79.77</b> $\pm$ 0.50 |
|          |       | COFFEQA | <b>79.07</b> $\pm$ 0.19 | 79.30 $\pm$ 0.59        | <b>79.91</b> $\pm$ 0.25 | <b>80.28</b> $\pm$ 0.21 | 79.77 $\pm$ 0.49        | 79.67 $\pm$ 0.48        |
|          | F1    | KD      | 86.09 $\pm$ 0.15        | 86.50 $\pm$ 0.51        | 86.80 $\pm$ 0.45        | 87.19 $\pm$ 0.14        | <b>87.24</b> $\pm$ 0.41 | 86.76 $\pm$ 0.48        |
|          |       | COFFEQA | <b>86.17</b> $\pm$ 0.14 | <b>86.92</b> $\pm$ 0.28 | <b>86.95</b> $\pm$ 0.25 | <b>87.28</b> $\pm$ 0.29 | 86.87 $\pm$ 0.50        | <b>86.84</b> $\pm$ 0.41 |
| NewsQA   | EM    | KD      | 59.65 $\pm$ 0.78        | 63.53 $\pm$ 1.78        | 67.24 $\pm$ 1.71        | <b>68.25</b> $\pm$ 2.46 | <b>70.94</b> $\pm$ 1.83 | 65.92 $\pm$ 4.40        |
|          |       | COFFEQA | <b>64.69</b> $\pm$ 1.43 | <b>64.11</b> $\pm$ 1.36 | <b>67.26</b> $\pm$ 1.82 | 68.05 $\pm$ 2.15        | 69.26 $\pm$ 1.10        | <b>66.67</b> $\pm$ 2.20 |
|          | F1    | KD      | 70.17 $\pm$ 0.66        | 73.11 $\pm$ 1.47        | <b>76.02</b> $\pm$ 1.13 | <b>76.41</b> $\pm$ 1.55 | <b>78.21</b> $\pm$ 1.14 | 74.78 $\pm$ 3.16        |
|          |       | COFFEQA | <b>73.81</b> $\pm$ 0.91 | <b>73.52</b> $\pm$ 1.18 | 75.80 $\pm$ 1.39        | 76.35 $\pm$ 1.50        | 77.44 $\pm$ 0.98        | <b>75.38</b> $\pm$ 1.68 |
| TriviaQA | EM    | KD      | 78.66 $\pm$ 0.10        | 75.58 $\pm$ 2.67        | <b>73.19</b> $\pm$ 0.95 | <b>72.53</b> $\pm$ 0.77 | 72.75 $\pm$ 1.31        | <b>74.54</b> $\pm$ 2.60 |
|          |       | COFFEQA | <b>78.86</b> $\pm$ 0.12 | <b>76.45</b> $\pm$ 1.50 | 72.80 $\pm$ 0.91        | 71.47 $\pm$ 1.50        | <b>72.90</b> $\pm$ 0.95 | 74.50 $\pm$ 3.06        |
|          | F1    | KD      | 85.79 $\pm$ 0.10        | 83.72 $\pm$ 1.82        | <b>81.93</b> $\pm$ 0.86 | 80.52 $\pm$ 1.04        | 81.43 $\pm$ 0.97        | 82.68 $\pm$ 2.09        |
|          |       | COFFEQA | <b>85.90</b> $\pm$ 0.08 | <b>84.09</b> $\pm$ 1.16 | 81.71 $\pm$ 0.66        | <b>81.08</b> $\pm$ 0.60 | <b>81.62</b> $\pm$ 0.66 | <b>82.88</b> $\pm$ 2.05 |
| HotpotQA | EM    | KD      | 73.97 $\pm$ 0.95        | 72.55 $\pm$ 1.43        | 71.68 $\pm$ 1.08        | <b>74.26</b> $\pm$ 1.72 | <b>74.24</b> $\pm$ 2.44 | 73.34 $\pm$ 1.17        |
|          |       | COFFEQA | <b>75.34</b> $\pm$ 0.50 | <b>73.55</b> $\pm$ 0.87 | <b>74.08</b> $\pm$ 1.57 | 73.43 $\pm$ 1.17        | 73.39 $\pm$ 1.69        | <b>73.96</b> $\pm$ 0.82 |
|          | F1    | KD      | 81.87 $\pm$ 0.57        | <b>81.33</b> $\pm$ 0.65 | 79.52 $\pm$ 0.75        | 80.79 $\pm$ 0.86        | <b>81.52</b> $\pm$ 1.97 | 81.01 $\pm$ 0.92        |
|          |       | COFFEQA | <b>82.50</b> $\pm$ 0.62 | 80.65 $\pm$ 0.80        | <b>81.36</b> $\pm$ 0.81 | <b>82.20</b> $\pm$ 0.56 | 80.96 $\pm$ 1.40        | <b>81.53</b> $\pm$ 0.79 |
| AVG      | EM    | KD      | 70.89 $\pm$ 8.96        | 70.92 $\pm$ 7.41        | 70.98 $\pm$ 6.25        | <b>71.85</b> $\pm$ 6.09 | <b>72.55</b> $\pm$ 5.66 | 71.44 $\pm$ 0.74        |
|          |       | COFFEQA | <b>72.18</b> $\pm$ 7.81 | <b>71.47</b> $\pm$ 7.10 | <b>71.54</b> $\pm$ 6.30 | 71.70 $\pm$ 5.73        | 71.95 $\pm$ 5.66        | <b>71.77</b> $\pm$ 0.29 |
|          | F1    | KD      | 78.37 $\pm$ 8.71        | 78.55 $\pm$ 7.69        | 78.51 $\pm$ 6.93        | 79.00 $\pm$ 6.29        | <b>79.51</b> $\pm$ 6.65 | 78.79 $\pm$ 0.47        |
|          |       | COFFEQA | <b>79.24</b> $\pm$ 8.11 | <b>78.74</b> $\pm$ 7.59 | <b>78.97</b> $\pm$ 6.82 | <b>79.17</b> $\pm$ 6.91 | 79.33 $\pm$ 6.32        | <b>79.09</b> $\pm$ 0.24 |

on reasoning over the given context. Across our experiments, we observe stronger performance on counterfactual test sets, indicating that our approach effectively improves robustness under knowledge conflict. More details can be found in Table 5 and 6 in Appendix D.1.

**Low CFD counterfactual setting perform best among CFD variants.** To validate the effectiveness of the **Context Following Difficulty (CFD)** metric, we conduct a comprehensive ablation study across four distinct configurations, applying the metric to both factual and counterfactual samples. Although CFD was originally conceptualized to evaluate counterfactual transitions, it is equally applicable to factual samples. In this setting, a low CFD score identifies factual instances that most accurately generates the expected answer  $A$  using the given context  $C$ . For evaluation, we replace the  $\mathcal{D}_{train}$  training set in Algorithm 1 with four variants: (1) **Lowest CFD Factuals**, (2) **Highest CFD Factuals**, (3) **Lowest CFD Counterfactuals**, and (4) **Highest CFD Counterfactuals**. Experiments are performed across five in-domain benchmarks for all  $k$ -shot settings, with results reported as the average across  $k$ -shots per dataset. Notably, the configuration utilizing the **lowest CFD counterfactuals** achieved superior performance among all four settings, confirming that selecting low CFD counterfactuals, which have high teacher-model alignment, is critical for effective knowledge distillation. Detailed results are provided in Table 7 of the Appendix D.2, respectively.

## 5 Conclusion

In this paper, we present a counterfactual-guided knowledge distillation method for open-book question answering in the few-shot setting. Our approach leverages counterfactuals to force the student model to mimic the teacher model’s behavior in the presence of knowledge conflicts. We introduce a new metric, Context Following Difficulty (CFD), to identify counterfactuals that are well suited for transferring teacher capabilities to the student. Extensive experiments across multiple datasets and LLMs demonstrate the effectiveness of our method, highlighting its scalability across different model sizes and its data efficiency in few-shot settings. Furthermore, our method generalizes well on out-of-domain datasets. We discuss limitations and future work in Appendix A. Importantly, our findings suggest that explicitly modeling context sensitivity through counterfactuals is a promising direction for improving robustness in distillation for generative QA systems under few-shot setting.

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# Appendix

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## A Limitations and Future Work

The proposed framework for generating counterfactuals is currently limited to entity-based answer examples. Additionally, it does not incorporate feedback from the teacher model during counterfactual generation. This could be improved by integrating the teacher model into the generation process, potentially through LLM-based prompting strategies.

Furthermore, CFD could be directly incorporated into the counterfactual generation pipeline to guide the selection of higher-quality counterfactuals for each sample. Our method also assumes access to the full dataset during knowledge distillation. However, if high-quality counterfactuals can be generated for all samples, it may be sufficient to randomly sample a subset of factual examples, generate their corresponding counterfactuals, and perform distillation without depending on the availability of the full dataset.

Finally, the current approach is limited to open-book QA. Extending the definition and construction of counterfactuals to other generative tasks would allow the framework to generalize beyond QA settings.

## B Experimental Setup Details

### B.1 Datasets

We used eight open-domain question answering benchmarks from the MRQA 2019 shared task [32] for our experiments. The MRQA datasets are designed for generalization in open-book question answering, where samples are drawn from diverse distributions. The datasets have following properties in common.

- **Question distribution:** The test sets contain different styles of questions, such as entity-based and relational questions, collected from diverse sources, including crowdworkers, domain experts, and exam writers.
- **Context distribution:** Passages are drawn from heterogeneous sources such as science articles, novels, medical abstracts, news articles.
- **Answerability:** The answer can be generated using the provided context, and the answer span appears within the first 800 tokens of the passage.

The training and validation sets for the teacher are drawn from the original training splits after removing duplicate entries. Dataset statistics are reported in Table 4. We generate counterfactuals only for the samples with entity-based answers. Statistics of the knowledge distillation datasets are also shown in Table 4, where *train pairs* denote factual-counterfactual training pairs. The in-domain test sets for knowledge distillation are taken from the original development splits after filtering for entity-based answer samples. The out-of-domain test set counts are as follows: BioASQ (115), DROP (496), and DuoRC (450).

Table 4: Statistics of teacher and student datasets used in our experiments. Teacher datasets contain the original training/validation splits, while student datasets contain paired factual-counterfactual instances for distillation and evaluation.

| Dataset  | Teacher Dataset |        | Student Dataset |            |            |
|----------|-----------------|--------|-----------------|------------|------------|
|          | Train           | Val.   | Train Pairs     | Val. Pairs | Test Pairs |
| NQ       | 93,663          | 10,408 | 34,383          | 3,821      | 2,284      |
| SQuAD    | 77,847          | 8,651  | 16,341          | 1,816      | 2,791      |
| NewsQA   | 66,655          | 7,407  | 10,956          | 1,218      | 691        |
| TriviaQA | 55,519          | 6,170  | 4,415           | 491        | 608        |
| HotpotQA | 65,630          | 7,294  | 28,956          | 3,218      | 2,053      |

## B.2 Fine-tuning

All fine-tuning experiments on T5-based models were conducted on a machine equipped with four NVIDIA RTX 6000 Ada GPUs. Experiments involving FLAN-T5 models were performed using a single NVIDIA A100 GPU. For all fine-tuning experiments, we used a batch size of 4, a learning rate of  $1 \times 10^{-4}$ , and a maximum of 5 training epochs. Training was stopped early based on validation loss using an early stopping criterion with a patience of 3 validation checks. On average, all models converged within 1 to 2 epochs.

## B.3 Implementation details of KD

Standard distillation loss was used to implement knowledge distillation with temperature set to 1. The weight for the weighted average between two loss terms ( $\alpha$ ) was set to 0.5. We used a batch size of 4 with a learning rate of  $1 \times 10^{-4}$ . The maximum number of training epochs was set to 100, with early stopping based on validation loss using a patience of 3 validation checks.

## C Proof of Theorem 1

In this section, we provide the formal derivation for the upper bound on the Hausdorff distance between the teacher’s context-response boundary  $\mathcal{M}_t$  and the student’s context-response boundary  $\mathcal{M}_s$ .

By definition, the Hausdorff distance  $d_H(\mathcal{M}_s, \mathcal{M}_t)$  is the maximum of two directed distances:

$$d_H(\mathcal{M}_t, \mathcal{M}_s) := \max \left\{ \sup_{c_t \in \mathcal{M}_t} \inf_{c_s \in \mathcal{M}_s} d(c_t, c_s), \sup_{c_s \in \mathcal{M}_s} \inf_{c_t \in \mathcal{M}_t} d(c_t, c_s) \right\} \quad (4)$$

We bound these two components individually [63].

### Part 1: Bounding the deviation from $\mathcal{M}_t$ to $\mathcal{M}_s$

Let  $t \in \mathcal{M}_t$  be an arbitrary point on the teacher’s boundary. Based on the assumption 3, the context-sensitivity space has an  $\epsilon$ -spread. Therefore, there exists a teacher crossing point  $t_i^* \in \mathcal{M}_t$  belonging to a counterfactual pair  $(c_i, c'_i)$  such that (see Figure 5):

$$\|t - t_i^*\| \leq \epsilon \quad (5)$$

Under assumption 1, the perturbation between the factual and counterfactual contexts is bounded by  $\|c_i - c'_i\| \leq \alpha$ . Given the exact distillation assumption 2, the student’s boundary  $\mathcal{M}_s$  must intersect the line segment  $[c_i, c'_i]$  at some point  $s_i^*$ . Because both  $t_i^*$  and  $s_i^*$  reside on the same segment of maximum length  $\alpha$ , we have:

$$\|t_i^* - s_i^*\| \leq \|c_i - c'_i\| \leq \alpha \quad (6)$$

By applying the triangle inequality to the distance between the arbitrary teacher point  $t$  and the student boundary point  $s_i^*$ :

$$\|t - s_i^*\| \leq \|t - t_i^*\| + \|t_i^* - s_i^*\| \leq \epsilon + \alpha \quad (7)$$

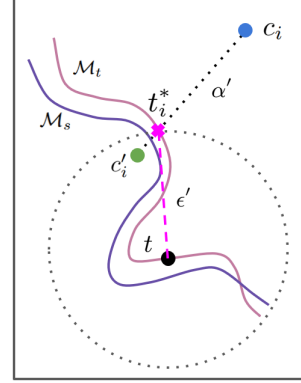


Figure 5: Visualization of the distances between  $\mathcal{M}_t$  and  $\mathcal{M}_s$ .  $c_i$  and  $c'_i$  are factual and counterfactual pairs with  $\|c_i - c'_i\| = \alpha' \leq \alpha$ .  $t$  is an arbitrary point in  $\mathcal{M}_t$  that finds some  $t_i^*$  cut point in  $\mathcal{M}_t$  within  $\epsilon'$  distance ( $\epsilon' \leq \epsilon$ ).

Since there exists at least one point  $s_i^* \in \mathcal{M}_s$  within distance  $\epsilon + \alpha$  of  $t$ , the infimum distance satisfies:

$$\inf_{s \in \mathcal{M}_s} \|t - s\| \leq \epsilon + \alpha \quad (8)$$

Taking the supremum over all points  $t \in \mathcal{M}_t$  yields:

$$\sup_{t \in \mathcal{M}_t} \inf_{s \in \mathcal{M}_s} \|c - s\| \leq \epsilon + \alpha \quad (9)$$

## Part 2: Bounding the deviation from $\mathcal{M}_s$ to $\mathcal{M}_t$

We follow a symmetric logic for the student boundary. For any point  $s \in \mathcal{M}_s$ , assumption 3 guarantees the existence of a student boundary cutpoint  $s_i^*$  such that  $\|s - s_i^*\| \leq \epsilon$ .

From the relationship established in Part 1, the distance between this student cutpoint and the corresponding teacher cutpoint  $t_i^*$  is  $\|s_i^* - t_i^*\| \leq \alpha$ . Applying the triangle inequality:

$$\|s - t_i^*\| \leq \|s - s_i^*\| + \|s_i^* - t_i^*\| \leq \epsilon + \alpha \quad (10)$$

Since  $t_i^* \in \mathcal{M}_t$ , it follows that  $\inf_{t \in \mathcal{M}_t} \|s - c\| \leq \epsilon + \alpha$ . Taking the supremum over all  $s \in \mathcal{M}_s$ :

$$\sup_{s \in \mathcal{M}_s} \inf_{t \in \mathcal{M}_t} \|s - c\| \leq \epsilon + \alpha \quad (11)$$

## Final Result

Combining the bounds from the two directed distances:

$$d_H(\mathcal{M}_s, \mathcal{M}_t) \leq \max\{\epsilon + \alpha, \epsilon + \alpha\} = \alpha + \epsilon \quad (12)$$

This confirms that the student boundary remains within an  $(\alpha + \epsilon)$ -tube of the teacher boundary.

## D Ablation Study Results

### D.1 Performance on Counterfactual Datasets

Table 5: Performance on in-domain counterfactual test sets using the T5 model. Bold values indicate the better performing method (KD or COFFEQA). Blue shading highlights aggregate results.

| Dataset  | Score | Method  | Total Samples ( $k$ )    |                          |                          |                          |                          | Avg $\pm$ std           |
|----------|-------|---------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------|
|          |       |         | 8                        | 16                       | 32                       | 64                       | 128                      |                         |
| NQ       | EM    | KD      | <b>53.83</b> $\pm$ 2.07  | 52.98 $\pm$ 2.51         | 51.42 $\pm$ 4.34         | 54.13 $\pm$ 1.13         | 51.95 $\pm$ 4.10         | 52.86 $\pm$ 1.17        |
|          |       | COFFEQA | 52.71 $\pm$ 2.86         | <b>54.94</b> $\pm$ 1.03  | <b>55.27</b> $\pm$ 0.35  | <b>54.44</b> $\pm$ 2.93  | <b>54.70</b> $\pm$ 4.43  | <b>54.41</b> $\pm$ 1.00 |
|          | F1    | KD      | <b>57.62</b> $\pm$ 1.09  | 56.80 $\pm$ 1.85         | 55.62 $\pm$ 3.38         | 57.53 $\pm$ 0.89         | 55.94 $\pm$ 3.35         | 56.70 $\pm$ 0.91        |
|          |       | COFFEQA | 56.76 $\pm$ 1.87         | <b>58.51</b> $\pm$ 0.88  | <b>58.91</b> $\pm$ 0.52  | <b>58.38</b> $\pm$ 2.32  | <b>58.52</b> $\pm$ 3.64  | <b>58.22</b> $\pm$ 0.84 |
| SQuAD    | EM    | KD      | 78.44 $\pm$ 0.54         | 77.46 $\pm$ 0.26         | 77.61 $\pm$ 0.72         | 76.42 $\pm$ 2.46         | 77.16 $\pm$ 1.59         | 77.42 $\pm$ 0.73        |
|          |       | COFFEQA | <b>78.78</b> $\pm$ 0.66  | <b>78.65</b> $\pm$ 0.46  | <b>78.01</b> $\pm$ 0.83  | <b>79.23</b> $\pm$ 0.91  | <b>78.52</b> $\pm$ 2.62  | <b>78.64</b> $\pm$ 0.44 |
|          | F1    | KD      | 85.22 $\pm$ 0.37         | 84.68 $\pm$ 0.08         | <b>84.70</b> $\pm$ 0.64  | 83.29 $\pm$ 2.87         | 84.55 $\pm$ 1.13         | 84.49 $\pm$ 0.72        |
|          |       | COFFEQA | <b>85.38</b> $\pm$ 0.43  | <b>85.23</b> $\pm$ 0.50  | 84.39 $\pm$ 0.46         | <b>85.62</b> $\pm$ 0.46  | <b>84.98</b> $\pm$ 1.64  | <b>85.12</b> $\pm$ 0.47 |
| NewsQA   | EM    | KD      | 59.45 $\pm$ 1.90         | <b>62.26</b> $\pm$ 1.17  | 63.45 $\pm$ 2.00         | <b>65.13</b> $\pm$ 1.58  | 66.46 $\pm$ 0.94         | 63.35 $\pm$ 2.70        |
|          |       | COFFEQA | <b>60.96</b> $\pm$ 1.98  | 60.87 $\pm$ 2.62         | <b>63.48</b> $\pm$ 1.64  | 64.55 $\pm$ 0.78         | <b>67.19</b> $\pm$ 1.25  | <b>63.41</b> $\pm$ 2.65 |
|          | F1    | KD      | 65.54 $\pm$ 2.21         | <b>67.54</b> $\pm$ 1.14  | 68.71 $\pm$ 1.62         | 69.91 $\pm$ 1.84         | 70.88 $\pm$ 0.68         | 68.52 $\pm$ 2.08        |
|          |       | COFFEQA | <b>67.63</b> $\pm$ 1.65  | 67.19 $\pm$ 2.34         | <b>69.42</b> $\pm$ 1.36  | <b>70.14</b> $\pm$ 0.70  | <b>72.20</b> $\pm$ 1.87  | <b>69.32</b> $\pm$ 2.02 |
| TriviaQA | EM    | KD      | 49.03 $\pm$ 1.05         | 48.96 $\pm$ 2.24         | <b>57.00</b> $\pm$ 2.95  | <b>62.41</b> $\pm$ 3.28  | 63.43 $\pm$ 1.42         | 56.17 $\pm$ 6.99        |
|          |       | COFFEQA | <b>50.77</b> $\pm$ 0.55  | <b>50.48</b> $\pm$ 1.94  | 52.92 $\pm$ 2.27         | 60.16 $\pm$ 2.20         | <b>67.28</b> $\pm$ 0.73  | <b>56.32</b> $\pm$ 7.27 |
|          | F1    | KD      | <b>56.19</b> $\pm$ 0.70  | <b>55.81</b> $\pm$ 1.99  | <b>60.39</b> $\pm$ 3.01  | <b>65.00</b> $\pm$ 2.58  | 65.07 $\pm$ 1.70         | <b>60.49</b> $\pm$ 4.52 |
|          |       | COFFEQA | 55.15 $\pm$ 0.93         | 54.54 $\pm$ 2.07         | 58.08 $\pm$ 1.63         | 63.31 $\pm$ 1.49         | <b>69.26</b> $\pm$ 0.66  | 60.07 $\pm$ 6.20        |
| HotpotQA | EM    | KD      | <b>52.89</b> $\pm$ 1.01  | <b>53.74</b> $\pm$ 0.60  | 53.30 $\pm$ 1.79         | 52.95 $\pm$ 1.74         | <b>55.99</b> $\pm$ 2.57  | <b>53.77</b> $\pm$ 1.28 |
|          |       | COFFEQA | 51.62 $\pm$ 0.45         | 53.20 $\pm$ 1.12         | <b>53.55</b> $\pm$ 0.80  | <b>54.67</b> $\pm$ 0.64  | 54.49 $\pm$ 1.02         | 53.51 $\pm$ 1.22        |
|          | F1    | KD      | <b>58.43</b> $\pm$ 0.85  | <b>59.00</b> $\pm$ 0.44  | 58.29 $\pm$ 1.64         | 58.02 $\pm$ 1.93         | <b>61.11</b> $\pm$ 2.73  | <b>58.97</b> $\pm$ 1.25 |
|          |       | COFFEQA | 57.26 $\pm$ 0.31         | 58.55 $\pm$ 0.85         | <b>58.42</b> $\pm$ 0.54  | <b>59.52</b> $\pm$ 0.69  | 59.04 $\pm$ 0.95         | 58.56 $\pm$ 0.85        |
| AVG      | EM    | KD      | 58.73 $\pm$ 11.63        | 59.08 $\pm$ 11.36        | 60.56 $\pm$ 10.58        | 62.21 $\pm$ 9.51         | 63.00 $\pm$ 9.80         | 60.71 $\pm$ 1.88        |
|          |       | COFFEQA | <b>58.97</b> $\pm$ 11.80 | <b>59.63</b> $\pm$ 11.30 | <b>60.65</b> $\pm$ 10.59 | <b>62.61</b> $\pm$ 10.20 | <b>64.44</b> $\pm$ 10.10 | <b>61.26</b> $\pm$ 2.25 |
|          | F1    | KD      | <b>64.60</b> $\pm$ 12.08 | 64.77 $\pm$ 12.05        | 65.54 $\pm$ 11.78        | 66.75 $\pm$ 10.58        | 67.51 $\pm$ 10.98        | 65.83 $\pm$ 1.26        |
|          |       | COFFEQA | 64.44 $\pm$ 12.70        | <b>64.80</b> $\pm$ 12.32 | <b>65.84</b> $\pm$ 11.40 | <b>67.39</b> $\pm$ 11.18 | <b>68.80</b> $\pm$ 10.89 | <b>66.26</b> $\pm$ 1.83 |

Table 6: Performance on in-domain counterfactual test sets using the FLAN-T5 model. Bold values indicate the better performing method (KD or COFFEQA). Blue shading highlights aggregate results.

| Dataset  | Score | Method  | Total Samples ( $k$ )    |                         |                         |                         |                         | Avg $\pm$ std           |
|----------|-------|---------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|          |       |         | 8                        | 16                      | 32                      | 64                      | 128                     |                         |
| NQ       | EM    | KD      | <b>58.76</b> $\pm$ 1.53  | <b>59.76</b> $\pm$ 0.91 | <b>60.47</b> $\pm$ 0.68 | 61.05 $\pm$ 1.45        | 61.01 $\pm$ 1.12        | 60.21 $\pm$ 0.96        |
|          |       | COFFEQA | 57.87 $\pm$ 1.70         | 59.20 $\pm$ 1.19        | 59.68 $\pm$ 2.85        | <b>62.28</b> $\pm$ 1.92 | <b>62.32</b> $\pm$ 1.46 | <b>60.27</b> $\pm$ 1.97 |
|          | F1    | KD      | <b>61.70</b> $\pm$ 1.01  | <b>62.85</b> $\pm$ 0.74 | <b>63.79</b> $\pm$ 0.70 | 64.25 $\pm$ 1.12        | 64.48 $\pm$ 0.72        | 63.41 $\pm$ 1.14        |
|          |       | COFFEQA | 61.24 $\pm$ 1.10         | 62.47 $\pm$ 0.66        | 63.20 $\pm$ 2.22        | <b>65.45</b> $\pm$ 1.64 | <b>65.56</b> $\pm$ 1.24 | <b>63.58</b> $\pm$ 1.89 |
| SQuAD    | EM    | KD      | 74.83 $\pm$ 1.01         | 77.05 $\pm$ 0.50        | 77.74 $\pm$ 1.01        | 78.43 $\pm$ 0.53        | 79.61 $\pm$ 0.99        | 77.53 $\pm$ 1.78        |
|          |       | COFFEQA | <b>76.65</b> $\pm$ 1.82  | <b>77.47</b> $\pm$ 1.26 | <b>78.29</b> $\pm$ 1.55 | <b>79.42</b> $\pm$ 1.01 | <b>80.34</b> $\pm$ 0.39 | <b>78.43</b> $\pm$ 1.48 |
|          | F1    | KD      | 81.02 $\pm$ 0.74         | 82.84 $\pm$ 0.54        | 83.59 $\pm$ 1.02        | 84.06 $\pm$ 0.39        | 85.28 $\pm$ 1.08        | 83.36 $\pm$ 1.58        |
|          |       | COFFEQA | <b>82.58</b> $\pm$ 1.55  | <b>83.50</b> $\pm$ 1.10 | <b>84.31</b> $\pm$ 1.22 | <b>85.64</b> $\pm$ 0.79 | <b>86.18</b> $\pm$ 0.47 | <b>84.44</b> $\pm$ 1.49 |
| NewsQA   | EM    | KD      | 53.08 $\pm$ 0.40         | 55.75 $\pm$ 1.43        | 58.12 $\pm$ 2.20        | 59.77 $\pm$ 2.69        | <b>61.42</b> $\pm$ 2.83 | 57.63 $\pm$ 3.29        |
|          |       | COFFEQA | <b>57.51</b> $\pm$ 1.17  | <b>56.85</b> $\pm$ 0.75 | <b>59.80</b> $\pm$ 1.73 | <b>59.80</b> $\pm$ 1.83 | 60.64 $\pm$ 1.61        | <b>58.92</b> $\pm$ 1.64 |
|          | F1    | KD      | 62.92 $\pm$ 0.37         | 65.01 $\pm$ 1.02        | 66.30 $\pm$ 1.77        | 67.45 $\pm$ 2.16        | <b>68.47</b> $\pm$ 2.05 | 66.03 $\pm$ 2.17        |
|          |       | COFFEQA | <b>66.20</b> $\pm$ 0.80  | <b>65.73</b> $\pm$ 0.59 | <b>67.50</b> $\pm$ 0.97 | <b>67.70</b> $\pm$ 1.24 | 68.36 $\pm$ 1.26        | <b>67.10</b> $\pm$ 1.09 |
| TriviaQA | EM    | KD      | 72.35 $\pm$ 0.11         | 69.52 $\pm$ 1.96        | 69.17 $\pm$ 0.89        | 69.57 $\pm$ 1.08        | 71.11 $\pm$ 1.26        | 70.34 $\pm$ 1.35        |
|          |       | COFFEQA | <b>73.16</b> $\pm$ 0.31  | <b>70.52</b> $\pm$ 1.39 | <b>69.24</b> $\pm$ 0.74 | <b>70.31</b> $\pm$ 1.28 | 71.00 $\pm$ 1.18        | <b>70.85</b> $\pm$ 1.44 |
|          | F1    | KD      | 79.05 $\pm$ 0.12         | 76.79 $\pm$ 1.73        | 75.89 $\pm$ 0.92        | 75.85 $\pm$ 0.84        | 77.26 $\pm$ 1.31        | 76.97 $\pm$ 1.31        |
|          |       | COFFEQA | <b>79.70</b> $\pm$ 0.30  | <b>77.15</b> $\pm$ 1.25 | <b>76.14</b> $\pm$ 0.69 | <b>76.87</b> $\pm$ 1.17 | <b>77.59</b> $\pm$ 1.12 | <b>77.49</b> $\pm$ 1.34 |
| HotpotQA | EM    | KD      | 56.27 $\pm$ 0.50         | <b>57.66</b> $\pm$ 0.74 | 57.23 $\pm$ 0.39        | 57.41 $\pm$ 1.29        | 56.50 $\pm$ 2.41        | 57.01 $\pm$ 0.60        |
|          |       | COFFEQA | <b>57.55</b> $\pm$ 0.65  | 56.53 $\pm$ 0.94        | <b>57.85</b> $\pm$ 0.72 | <b>58.08</b> $\pm$ 0.95 | <b>58.60</b> $\pm$ 1.02 | <b>57.72</b> $\pm$ 0.77 |
|          | F1    | KD      | 60.35 $\pm$ 0.55         | <b>61.91</b> $\pm$ 0.84 | 61.39 $\pm$ 0.44        | 61.87 $\pm$ 1.17        | 61.00 $\pm$ 2.25        | 61.30 $\pm$ 0.65        |
|          |       | COFFEQA | <b>62.01</b> $\pm$ 0.47  | 60.68 $\pm$ 0.95        | <b>62.25</b> $\pm$ 0.72 | <b>62.46</b> $\pm$ 0.86 | <b>62.76</b> $\pm$ 1.02 | <b>62.03</b> $\pm$ 0.80 |
| AVG      | EM    | KD      | 63.06 $\pm$ 10.13        | 63.95 $\pm$ 9.04        | 64.55 $\pm$ 9.04        | 65.25 $\pm$ 8.76        | 65.93 $\pm$ 9.32        | 64.55 $\pm$ 1.12        |
|          |       | COFFEQA | <b>64.55</b> $\pm$ 9.54  | <b>64.11</b> $\pm$ 9.41 | <b>64.97</b> $\pm$ 8.68 | <b>65.98</b> $\pm$ 8.86 | <b>66.58</b> $\pm$ 9.03 | <b>65.24</b> $\pm$ 1.02 |
|          | F1    | KD      | 69.01 $\pm$ 10.13        | 69.88 $\pm$ 9.39        | 70.19 $\pm$ 9.29        | 70.70 $\pm$ 9.15        | 71.30 $\pm$ 9.89        | 70.21 $\pm$ 0.86        |
|          |       | COFFEQA | <b>70.35</b> $\pm$ 10.08 | <b>69.91</b> $\pm$ 9.94 | <b>70.68</b> $\pm$ 9.39 | <b>71.62</b> $\pm$ 9.51 | <b>72.09</b> $\pm$ 9.65 | <b>70.93</b> $\pm$ 0.91 |

## D.2 CFD variants

Here, we examine the role of **Context Following Difficulty (CFD)** through a set of controlled experiments. First, we compute CFD scores for both factual and counterfactual samples. We then replace the  $\mathcal{D}_{train}$  training set in Algorithm 1 with four variants: (1) **Lowest CFD Factuals (L-CFD-fac)**, (2) **Highest CFD Factuals (H-CFD-fac)**, (3) **Lowest CFD Counterfactuals (L-CFD-cf)**, and (4) **Highest CFD Counterfactuals (H-CFD-cf)**.

Table 7: Performance comparison on in-domain benchmarks for different CFD variants using the T5 model. Bold values indicate better performance if our method performs well under the corresponding CFD variant method. Blue shading highlights average results across the CFD variant.

| CFD variant | Score | Method | NQ           | SQuAD        | NewsQA       | TriviaQA     | HotpotQA     | Avg $\pm$ std            |
|-------------|-------|--------|--------------|--------------|--------------|--------------|--------------|--------------------------|
| L-CFD-fac   | EM    | KD     | 55.49        | 78.13        | 69.61        | 56.04        | 64.53        | 64.76 $\pm$ 9.54         |
|             |       | our    | 54.62        | 78.06        | 67.17        | 55.74        | 62.54        | 63.63 $\pm$ 9.56         |
|             | F1    | KD     | 60.21        | 86.44        | 75.67        | 60.72        | 73.58        | 71.32 $\pm$ 11.05        |
|             |       | our    | 59.60        | 86.25        | 74.03        | <b>60.82</b> | 71.83        | 70.51 $\pm$ 10.90        |
| H-CFD-fac   | EM    | KD     | 53.41        | 77.00        | 69.06        | 55.93        | 65.87        | 64.25 $\pm$ 9.68         |
|             |       | our    | <b>54.31</b> | 76.39        | 64.59        | 55.38        | 65.81        | 63.30 $\pm$ 8.98         |
|             | F1    | KD     | 58.44        | 85.42        | 75.35        | 60.32        | 73.17        | 70.54 $\pm$ 11.21        |
|             |       | our    | 58.35        | 84.75        | 71.71        | 60.05        | 73.12        | 69.60 $\pm$ 10.77        |
| L-CFD-cf    | EM    | KD     | 55.53        | 77.33        | 69.18        | 55.25        | 61.07        | 63.67 $\pm$ 9.50         |
|             |       | our    | <b>56.53</b> | <b>78.16</b> | 68.99        | <b>56.89</b> | <b>63.62</b> | <b>64.84</b> $\pm$ 9.06  |
|             | F1    | KD     | 60.48        | 85.58        | 75.73        | 59.74        | 70.49        | 70.40 $\pm$ 10.85        |
|             |       | our    | <b>61.13</b> | <b>86.27</b> | 75.64        | <b>61.65</b> | <b>72.58</b> | <b>71.45</b> $\pm$ 10.50 |
| H-CFD-cf    | EM    | KD     | 55.00        | 76.88        | 60.82        | 55.63        | 67.92        | 63.25 $\pm$ 9.21         |
|             |       | our    | 51.13        | 75.94        | <b>61.45</b> | 55.60        | 67.45        | 62.31 $\pm$ 9.78         |
|             | F1    | KD     | 58.61        | 85.02        | 68.70        | 61.50        | 75.95        | 69.96 $\pm$ 10.78        |
|             |       | our    | 55.12        | 84.13        | 69.06        | 60.20        | <b>76.02</b> | 68.91 $\pm$ 11.71        |

As shown in Table 7, experiments are conducted across five in-domain benchmarks for all  $k$ -shot settings, with results reported as the average across  $k$ -shot configurations per dataset using the T5 model. Results are

reported using a single random seed, as the observed trends are consistent across multiple seeds. Notably, the configuration using **lowest CFD counterfactuals** consistently achieves superior performance across CFD variants, empirically confirming that selecting low CFD counterfactuals gives higher teacher-model alignment during distillation. Overall, the low-CFD counterfactual setting performs well on 4 out of 5 benchmarks.

## E OOD results of FLAN-T5

Table 8: Performance comparison on out-of-domain benchmarks using the Flan-T5 model. Bold values indicate the better performing method (KD or COFFEQA). Blue shading highlights aggregate results.

| Dataset | Score | Method  | Total Samples ( $k$ )    |                          |                          |                          |                          | Avg $\pm$ std           |
|---------|-------|---------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------|
|         |       |         | 8                        | 16                       | 32                       | 64                       | 128                      |                         |
| BioASQ  | EM    | KD      | 39.30 $\pm$ 1.29         | <b>41.91</b> $\pm$ 1.43  | 41.22 $\pm$ 1.58         | <b>42.09</b> $\pm$ 0.78  | <b>41.22</b> $\pm$ 1.46  | <b>41.15</b> $\pm$ 1.11 |
|         |       | COFFEQA | <b>40.52</b> $\pm$ 1.32  | 40.70 $\pm$ 1.13         | <b>41.39</b> $\pm$ 1.32  | 40.00 $\pm$ 1.74         | 40.17 $\pm$ 1.43         | 40.56 $\pm$ 0.54        |
|         | F1    | KD      | 63.49 $\pm$ 0.74         | <b>66.22</b> $\pm$ 1.27  | 64.69 $\pm$ 2.44         | <b>66.71</b> $\pm$ 0.85  | <b>65.78</b> $\pm$ 1.25  | 65.38 $\pm$ 1.29        |
|         |       | COFFEQA | <b>65.42</b> $\pm$ 1.55  | 65.48 $\pm$ 0.96         | <b>66.00</b> $\pm$ 1.48  | 65.93 $\pm$ 0.32         | 64.38 $\pm$ 1.16         | <b>65.44</b> $\pm$ 0.65 |
| DROP    | EM    | KD      | <b>65.73</b> $\pm$ 0.32  | 65.73 $\pm$ 0.82         | 65.28 $\pm$ 0.79         | 65.13 $\pm$ 1.92         | <b>66.37</b> $\pm$ 0.63  | 65.65 $\pm$ 0.48        |
|         |       | COFFEQA | 65.08 $\pm$ 0.52         | <b>66.13</b> $\pm$ 0.43  | <b>66.01</b> $\pm$ 0.44  | <b>65.97</b> $\pm$ 1.12  | 65.77 $\pm$ 0.85         | <b>65.79</b> $\pm$ 0.42 |
|         | F1    | KD      | <b>70.35</b> $\pm$ 0.16  | 70.38 $\pm$ 0.85         | <b>70.41</b> $\pm$ 0.54  | 70.23 $\pm$ 1.19         | 69.99 $\pm$ 0.58         | <b>70.27</b> $\pm$ 0.17 |
|         |       | COFFEQA | 69.82 $\pm$ 0.52         | <b>70.42</b> $\pm$ 0.56  | 69.65 $\pm$ 0.71         | <b>70.41</b> $\pm$ 0.18  | <b>70.82</b> $\pm$ 0.67  | 70.22 $\pm$ 0.48        |
| DuoRC   | EM    | KD      | 48.67 $\pm$ 0.52         | 48.58 $\pm$ 0.43         | 47.64 $\pm$ 0.73         | 47.20 $\pm$ 0.76         | <b>48.04</b> $\pm$ 1.40  | 48.03 $\pm$ 0.62        |
|         |       | COFFEQA | <b>49.07</b> $\pm$ 0.29  | <b>48.62</b> $\pm$ 0.37  | <b>48.18</b> $\pm$ 0.97  | <b>48.67</b> $\pm$ 1.14  | 46.58 $\pm$ 1.45         | <b>48.22</b> $\pm$ 0.97 |
|         | F1    | KD      | <b>54.39</b> $\pm$ 0.49  | 54.32 $\pm$ 0.53         | 53.80 $\pm$ 0.47         | 53.38 $\pm$ 0.61         | <b>54.07</b> $\pm$ 0.99  | 53.99 $\pm$ 0.41        |
|         |       | COFFEQA | 54.15 $\pm$ 0.35         | <b>54.91</b> $\pm$ 0.14  | <b>54.06</b> $\pm$ 0.76  | <b>54.59</b> $\pm$ 0.97  | 53.08 $\pm$ 1.04         | <b>54.16</b> $\pm$ 0.69 |
| AVG     | EM    | KD      | 51.23 $\pm$ 13.40        | <b>52.07</b> $\pm$ 12.29 | 51.38 $\pm$ 12.46        | 51.47 $\pm$ 12.10        | <b>51.88</b> $\pm$ 13.01 | <b>51.61</b> $\pm$ 0.35 |
|         |       | COFFEQA | <b>51.56</b> $\pm$ 12.47 | 51.82 $\pm$ 13.01        | <b>51.86</b> $\pm$ 12.72 | <b>51.55</b> $\pm$ 13.22 | 50.84 $\pm$ 13.32        | 51.52 $\pm$ 0.41        |
|         | F1    | KD      | 62.74 $\pm$ 8.01         | <b>63.64</b> $\pm$ 8.34  | 62.97 $\pm$ 8.44         | 63.44 $\pm$ 8.89         | <b>63.28</b> $\pm$ 8.25  | 63.21 $\pm$ 0.36        |
|         |       | COFFEQA | <b>63.13</b> $\pm$ 8.08  | 63.60 $\pm$ 7.92         | <b>63.24</b> $\pm$ 8.15  | <b>63.64</b> $\pm$ 8.15  | 62.76 $\pm$ 8.98         | <b>63.27</b> $\pm$ 0.36 |

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